

TOTAL, Australia

WMO: 947750

Lat: 32.6297S

Lon: 151.5919E

Elev: 30

StdP: 100.97

Time Zone: 10.00 (AUS)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	4.8	-4.1	2.7	19.5	-2.1	3.2	17.3	11.1	16.2	9.8	15.0	0.8	280	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.5	36.8	21.1	34.2	20.8	32.2	20.5	23.8	31.0	23.1	29.6	22.5	28.2	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	16.8	25.6	21.4	16.2	24.9	20.8	15.6	24.3	71.4	31.0	68.7	29.8	66.1	28.0	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.1	7.7	6.6	1.5	41.2	0.8	2.4	1.0	42.9	0.5	44.3	0.1	45.6	-0.5	47.3		
(5)			1.1	25.1	0.9	0.8	0.5	25.7	0.0	26.2	-0.5	26.6	-1.2	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	24.2	23.3	21.5	18.3	15.0	12.8	12.1	13.1	16.0	18.5	20.8	22.4
	DBStd	4.93	3.02	2.93	2.57	2.22	2.15	1.90	1.93	2.26	2.75	3.15	3.36	3.12
	HDD10.0	6	0	0	0	0	0	2	3	1	0	0	0	0
	HDD18.3	796	0	1	3	26	107	167	194	163	81	38	12	4
	CDD10.0	2974	441	373	357	250	154	84	67	97	180	262	325	383
	CDD18.3	723	183	141	102	26	2	0	0	1	11	42	87	129
	CDH23.3	6621	1756	1157	679	179	19	1	1	16	181	514	899	1218
	CDH26.7	2754	853	483	207	38	1	0	0	1	48	196	393	534
Wind	WSAvg	2.6	2.9	2.6	2.3	1.9	2.2	2.4	2.7	2.8	2.8	2.7	2.8	2.9
Precipitation	PrecAvg	884	76	116	91	83	68	84	52	46	56	48	83	81
	PrecMax	1329	366	478	318	334	235	376	143	148	154	159	249	199
	PrecMin	619	0	0	15	5	0	4	5	0	4	0	13	18
	PrecStd	175	68	97	62	72	60	69	31	35	41	36	51	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	38.6	33.9	30.4	25.4	21.0	22.3	25.4	31.0	34.2	37.8	38.6
		MCWB	22.2	22.4	20.9	18.5	16.5	14.3	13.7	14.2	15.4	17.2	19.9	20.1
	2%	DB	36.9	34.6	31.1	27.3	23.3	19.1	19.9	22.6	27.8	31.4	33.9	34.9
		MCWB	21.9	22.3	21.2	18.4	15.5	13.4	12.3	13.3	15.2	16.9	19.3	20.4
	5%	DB	34.1	31.9	29.0	25.3	21.6	18.0	18.2	20.7	25.3	28.7	31.2	32.0
		MCWB	21.5	21.6	21.0	18.0	15.2	13.2	11.7	12.5	14.5	17.0	19.4	20.2
	10%	DB	31.6	29.3	27.0	23.6	20.1	16.9	16.8	19.0	22.9	26.0	28.3	29.6
		MCWB	21.7	21.3	20.4	17.6	14.5	12.7	11.5	11.9	13.9	16.7	19.0	20.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.9	25.1	23.6	20.9	18.7	16.4	16.9	17.1	18.2	20.9	23.1	23.9
		MCDB	33.1	31.9	29.0	24.4	21.4	18.5	20.0	21.6	23.8	26.8	30.8	31.9
	2%	WB	23.9	23.9	22.5	19.9	17.6	15.4	14.6	15.1	17.1	19.6	21.7	22.7
		MCDB	31.8	30.1	27.6	24.0	20.9	17.4	17.1	18.8	22.6	25.5	28.6	29.5
	5%	WB	23.2	23.0	21.8	19.2	16.6	14.7	13.3	14.1	16.2	18.7	20.7	21.8
		MCDB	30.3	28.2	26.4	23.4	19.9	16.5	16.0	18.6	22.0	24.6	27.2	28.1
	10%	WB	22.5	22.1	21.1	18.5	15.7	13.9	12.5	13.1	15.2	17.9	19.9	21.1
		MCDB	28.6	27.0	25.3	22.4	18.9	15.8	15.3	17.4	20.8	23.4	25.8	26.8
Mean Daily Temperature Range	5% DB	MDBR	11.5	10.5	10.1	10.6	10.7	8.7	9.9	11.8	12.8	12.9	12.0	11.7
		MCDBR	18.0	15.8	13.6	13.9	13.0	10.5	12.4	15.3	17.8	18.7	18.1	17.4
		MCWBR	5.0	4.9	4.9	5.9	6.4	5.9	6.6	7.2	6.8	6.7	5.8	5.3
	5% WB	MCDBR	13.4	11.5	11.0	11.0	10.7	8.0	9.9	12.6	13.9	14.0	13.6	13.1
		MCWBR	4.4	4.4	4.5	5.2	5.9	5.3	6.4	7.0	6.5	6.2	5.3	5.0
Clear-Sky Solar Irradiance	taub	0.396	0.387	0.356	0.335	0.315	0.309	0.300	0.308	0.344	0.363	0.391	0.385	
	taud	2.432	2.480	2.557	2.588	2.595	2.609	2.614	2.578	2.470	2.449	2.413	2.452	
	Ebn at Noon	943	931	927	895	864	844	872	911	923	945	942	957	
	Edh at Noon	122	113	98	85	75	70	72	84	105	115	124	121	
All-Sky Solar Radiation	RadAvg	6.57	5.76	4.86	3.92	3.12	2.49	2.90	3.83	5.03	5.74	6.15	6.61	
	RadStd	0.43	0.51	0.43	0.32	0.25	0.20	0.23	0.25	0.28	0.51	0.58	0.56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	+0.45	N/A	N/A	+0.81	N/A	N/A	-2	-84	+188	+102

Nomenclature: See separate page